

Week 13 Assignment — Monte Carlo Simulation

CE 310 | Due: Wednesday of Week 14 by 9:00 AM | Submit as .ipynb (Colab notebook)

The lab applied Monte Carlo to building energy cost uncertainty. This assignment applies the same framework to two new engineering scenarios — one structural/infrastructure (Track A) and one building/energy (Track B). Complete your assigned track.

How to Submit

Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
Run all cells	Runtime → Run all — wait until every cell shows a checkmark, no spinners
File format	Download as .ipynb — File → Download → Download .ipynb (not PDF, not .py)
Print lines	Do not edit or delete the <code>print(f"ANSWER_A1 = ...")</code> lines
Seed	Use <code>SEED = 99</code> as set in the template — do not change it
File name	<code>lastname_firstname_weekN.ipynb</code> e.g. <code>smith_jane_week13.ipynb</code>
Submission	Upload the .ipynb file to D2L Dropbox before the deadline

Setup

```
import numpy as np
import matplotlib.pyplot as plt

SEED = 99      # Different seed from the lab
N      = 10_000
rng    = np.random.default_rng(SEED)
```

Use `rng` (not `np.random.seed`) for all random sampling in this assignment.

Track A — CE: Annual Highway Maintenance Cost

A transportation engineer manages **80 km** of rural highway. Each year, the total maintenance cost depends on three uncertain quantities:

Input	Distribution	Interpretation
Potholes per km	Normal(3.5, 0.9)	Road condition varies by winter severity
Repair cost per pothole	Normal(380, 60) \$/pothole	Labor and material cost uncertainty
Overhead rate	Normal(0.14, 0.025)	Contractor overhead markup

Annual cost model:

```
annual_cost = 80 * potholes_per_km * cost_per_pothole * (1 + overhead_rate)
```

P1 (2 pts): Sample all three inputs for $N = 10,000$ scenarios. Compute `annual_cost` for each. Set:

```
ANSWER_P1_mean = int(round(annual_cost.mean(), 0))
ANSWER_P1_std = int(round(annual_cost.std(), 0))
print(f"ANSWER_P1_mean = {ANSWER_P1_mean}")
print(f"ANSWER_P1_std = {ANSWER_P1_std}")
```

P2 (2 pts): Compute the 10th and 90th percentiles (the 80% confidence interval):

```
ANSWER_P2_p10 = int(round(np.percentile(annual_cost, 10), 0))
ANSWER_P2_p90 = int(round(np.percentile(annual_cost, 90), 0))
print(f"ANSWER_P2_p10 = {ANSWER_P2_p10}")
print(f"ANSWER_P2_p90 = {ANSWER_P2_p90}")
```

P3 (2 pts): The district's annual maintenance budget is **\$115,000**. What is the probability that actual cost exceeds the budget?

```
ANSWER_P3_prob_over_budget = round(np.mean(annual_cost > 115_000), 4)
print(f"ANSWER_P3_prob_over_budget = {ANSWER_P3_prob_over_budget}")
```

P4 (2 pts): Run an OAT variance decomposition. For each input, hold the other two at their means and measure `var(annual_cost)`. Which input contributes the most variance? Set:

```
ANSWER_P4_dominant_input = "potholes_per_km" # or "cost_per_pothole" or "overhead_rate"
print(f"ANSWER_P4_dominant_input = {ANSWER_P4_dominant_input}")
```

P5 (4 pts — written): In 2–4 sentences, explain which input dominates cost uncertainty and what the district could do operationally to reduce cost variability. Reference at least two specific numbers from your simulation.

Track B — ArcE: Demand Response Savings

A building owner is evaluating a **demand response (DR) program**: during summer peak hours, they will reduce cooling by curtailing air supply. The annual savings depend on uncertain factors:

Input	Distribution	Interpretation
Summer peak hours	Normal(620, 50) hr/yr	Hours when utility charges peak rate
Curtailment effectiveness	Normal(0.18, 0.04) fraction	% cooling load shed during DR events
Peak electricity rate	Normal(0.22, 0.03) \$/kWh	Utility peak demand rate
Off-peak rate	Normal(0.10, 0.01) \$/kWh	Rate during non-DR hours

From the Week 10 regression, **average cooling during BH hours 74.1 kWh/hr**.

Annual DR savings model:

$\text{savings} = \text{peak_hours} \times (74.1 \times \text{curtailment_eff}) \times (\text{peak_rate} - \text{off_peak_rate})$

This is the cost avoided by shifting load from the peak rate to the off-peak rate.

P1 (2 pts):

```
ANSWER_P1_mean_savings = int(round(savings.mean(), 0))
ANSWER_P1_std_savings = int(round(savings.std(), 0))
print(f"ANSWER_P1_mean_savings = {ANSWER_P1_mean_savings}")
print(f"ANSWER_P1_std_savings = {ANSWER_P1_std_savings}")
```

P2 (2 pts): Compute P10 and P90 of annual savings:

```
ANSWER_P2_p10 = int(round(np.percentile(savings, 10), 0))
ANSWER_P2_p90 = int(round(np.percentile(savings, 90), 0))
print(f"ANSWER_P2_p10 = {ANSWER_P2_p10}")
print(f"ANSWER_P2_p90 = {ANSWER_P2_p90}")
```

P3 (2 pts): The DR program has a fixed participation cost of **\$800/year**. What is the probability that the savings exceed this cost (i.e., the program is economically beneficial)?

```
ANSWER_P3_prob_beneficial = round(np.mean(savings > 800), 4)
print(f"ANSWER_P3_prob_beneficial = {ANSWER_P3_prob_beneficial}")
```

P4 (2 pts): OAT variance decomposition across all four inputs. Which input contributes the most variance to **savings**?

```
ANSWER_P4_dominant_input = "peak_hours" # or "curtailment_eff" or "peak_rate" or "off_peak"
print(f"ANSWER_P4_dominant_input = {ANSWER_P4_dominant_input}")
```

P5 (4 pts — written): In 2–4 sentences, interpret the P10–P90 range for the building owner. Is participating in the DR program a low-risk or high-risk financial decision based on your results? What would you tell the owner?

Deliverable

Submit a single Colab notebook (.ipynb) with all cells executed. All ANSWER_ variables must appear in cell output. The written P5 response should be in a markdown cell immediately after your P4 code cell.